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PUBLIC COMMENT

Request for Information: Accelerating the Adoption and Use of
Artificial Intelligence as Part of Clinical Care

RFI Title	Accelerating the Adoption and Use of AI as Part of Clinical Care
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Introduction

Scientix.ai Inc. is a Delaware C-Corp specializing in data infrastructure for regulated industries like healthcare & life sciences. Our leadership includes veterans of Amazon Web Services (AWS) and the pharmaceutical sector who specialize in resolving the structural friction that prevents clinical organizations from extracting value from their data. We appreciate the opportunity to provide input on this RFI and respond below to the questions where our operational experience is most directly relevant.

Our perspective is grounded in a consistent observation across engagements: **the primary barriers to clinical AI adoption are infrastructural, not algorithmic.** The AI models themselves have reached impressive capability thresholds. What remains broken is the data substrate they depend on, the fragmented systems, informal workarounds, and the absence of deterministic validation layers that clinical AI requires to earn clinician trust.

Question #1: *What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?*

The biggest barriers to private-sector innovation are not algorithmic—they are infrastructural. Three systemic forces constrain AI adoption in clinical care:

1. The Compounding Cost of System Integration

Current “application-first” paradigms treat data as a secondary byproduct of siloed software. This creates a compounding integration burden where the complexity and cost of connecting disparate systems consume the capital and time intended for innovation. Research consistently finds that data workers spend over 40% of their time searching for, cleaning, and reconciling data rather than analyzing it.[1] Vendors often protect proprietary data structures, leading to lock-in and a fragmented ecosystem that is nearly impossible to maintain.

2. The Gap Between Documented Workflows and Operational Reality

Most AI implementations are built on how IT departments believe data flows (the documented workflow), which rarely reflects operational reality. Clinical and administrative staff frequently bypass official systems using unregulated spreadsheets and manual workarounds to bridge technical gaps. When AI models are trained on the official data pathway alone, they fail to generalize in real clinical environments—producing outputs that are technically plausible but operationally wrong.

3. The Trust Gap: Probabilistic Risk in Deterministic Care

Clinical care demands deterministic accountability—the ability to trace exactly which data informed a decision and verify that it was correct. Purely probabilistic AI (such as large language models) cannot provide this traceability on its own. In high-stakes environments, non-deterministic behavior creates a trust gap: hallucinations and inconsistent outputs are not merely inconvenient, they are potentially dangerous. Most current architectures lack a **deterministic validation layer**—a verifiable record of truth that enforces data integrity (exact timestamps, immutable records) before probabilistic insights are applied on top.

We address these barriers by helping organizations transition from siloed, application-centric architectures to context-first data infrastructure built on open standards (such as NGSI-LD), where institutional knowledge becomes a persistent, governed asset rather than a byproduct of any single vendor's software.

Question #2: *What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why?*

HHS should prioritize regulatory and programmatic shifts that address the integration burden and trust gap described above. This requires updating legacy frameworks that were designed for static software, not adaptive AI systems.

Regulatory Priorities

Modernize HIPAA for AI-era data governance (45 C.F.R. Parts 160 and 164): Current de-identification and re-identification rules under § 164.514 should be updated to facilitate real-time, automated data governance that prevents silos without compromising patient privacy. As AI systems require richer longitudinal datasets, the rules should clarify how organizations can responsibly prepare data for model training while maintaining HIPAA compliance.

Expand algorithmic transparency under HTI-1 (45 C.F.R. § 170.315(b)(11)): The HTI-1 Final Rule's transparency requirements for Predictive Decision Support Interventions are a strong foundation. HHS should expand these to include mandatory reporting of serious adverse events attributed to AI-driven clinical recommendations, building on the existing 31 source attribute disclosure requirements.

Establish deterministic validation safe harbors: Create "qualified AI" safe harbor principles for tools that meet rigorous, HHS-endorsed standards for transparency and deterministic validation. This would mitigate liability risks for clinicians using AI as intended and incentivize developers to build verifiable systems rather than opaque ones.

Payment Policy Reforms

Temporary Medicare payment codes for AI-enabled interventions: The traditional fee-for-service model disincentivizes AI adoption because efficiency-driven tools (those reducing hospitalizations or shortening visits) can lower short-term billable services. Temporary payment codes—similar to those proposed for rural hospital access—would allow AI-enabled clinical interventions to be reimbursed while evidence of long-term value accumulates.

Strengthen human-in-the-loop requirements for medical necessity (42 C.F.R. § 422.101(c)): CMS's 2024 FAQs clarified that Medicare Advantage medical necessity determinations cannot rely solely on algorithmic outputs without considering individual patient circumstances. HHS should codify this principle more explicitly, ensuring that automated prior authorization systems include meaningful clinical review rather than defaulting to algorithmic denials.

Programmatic Design

Expand shared infrastructure through OneHHS: HHS should build shared, cloud-agnostic infrastructure that allows agencies like the FDA, CDC, and CMS to reuse validated AI components and standardized data models. This reduces duplicative investment and lowers the barrier to entry for smaller innovators.

Fund operational readiness assessments before implementation: R&D programs should incentivize organizations to map their actual operational workflows—including informal workarounds and data gaps—before committing federal capital to AI software procurement. Without this diagnostic step, AI implementations are built on faulty assumptions about how data actually flows.

Summary of Priority CFR Revisions

Regulation	Citation	Recommended Change
HIPAA Security Rule	45 C.F.R. § 164.308	Update to include AI-specific risk analysis and data-lineage requirements.
Health IT Certification (HTI-1)	45 C.F.R. § 170.315(b)(11)	Expand transparency requirements to include mandatory adverse event reporting for AI-driven clinical recommendations.
Medicare Advantage	42 C.F.R. § 422.101(c)	Codify the requirement that algorithmic prior authorizations include human review of individual clinical circumstances.
HIPAA Privacy Rule	45 C.F.R. § 164.514	Clarify de-identification standards for AI training datasets, enabling responsible real-time data governance.

Question #6: ***Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short?***

Where AI Has Exceeded Expectations

Medical imaging and triage: AI has significantly outperformed expectations in diagnostic imaging. A peer-reviewed Cedars-Sinai study found that AI-augmented radiological triage reduced hospital length of stay for pulmonary embolism by 26.3%.[3] AI-driven triage of patient portal messages reduced median clinician review time for high-acuity content from 22 hours to approximately 5 hours.[4]

Administrative automation: Automated billing and scheduling represent the fastest-growing use case. In 2026, 98% of surveyed health executives expect at least 10% cost savings from AI, with 37% targeting savings above 20%.[2] Early adopters of agentic AI—systems that coordinate tasks across billing, care delivery, and payments—report disproportionate productivity gains.[8]

Ambient documentation: Generative AI scribes have reduced the documentation burden on clinicians, directly addressing one of the primary drivers of provider burnout.[9]

Where AI Has Fallen Short

Shallow clinical reasoning: Many tools optimize for drafting notes rather than improving decisions. They frequently lack the longitudinal context (historical diagnoses, labs, patterns across encounters) needed for effective value-based care.

Correction burden on clinicians: Clinicians report spending up to three hours per week correcting or bypassing AI outputs due to missing context or misinterpreted patient data.[5] This erodes the efficiency gains that AI was intended to deliver.

Bias and evaluation gaps: While 65% of hospitals use predictive models, less than half evaluate these tools for potential bias.[6] Off-the-shelf products may not reflect local patient populations, creating a quiet disparity in care quality.

Hallucination and misinformation: Healthcare-specific AI models still maintain hallucination rates estimated at 10–30%. Research published in *The Lancet Digital Health* in February 2026 revealed that leading language models can repeat false medical claims as truth when the misinformation is wrapped in realistic clinical language.[7]

Novel Tools with Greatest Potential

The greatest potential lies in shifting from passive automation to context-aware, agentic systems: AI that autonomously gathers documentation from multiple sources, validates it against clinical requirements, and presents verified recommendations—rather than AI that simply generates text for human review. Equally promising are predictive remote monitoring tools (AI-powered wearables detecting early warning signs before escalation) and infrastructure layers that enforce data integrity at the point of creation, ensuring AI always reasons from verified facts rather than fragmented snapshots.

Question #7: ***Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles?***

While clinical AI champions, C-suite leaders (CIO/CDO/CEO), and multidisciplinary AI committees each play roles in adoption decisions, our experience across regulated environments reveals a more fundamental constraint that shapes all of these: the **executive digital fluency gap.**

The Leadership Fluency Gap

The most profound and overlooked administrative hurdle in 2026 is not technical—it is the gap between what organizational leaders *think* AI is and what it actually requires to succeed. While the vast majority of healthcare organizations recognize AI's importance, only about half of IT leaders feel confident in their implementation abilities. This stems from an executive mental model that treats AI as a passive software tool (purchase it, install it, benefit from it) rather than an adaptive system that requires ongoing governance, data infrastructure, and organizational redesign.

This fluency gap creates specific, observable pathologies:

Perpetual pilot programs: Leaders lacking digital fluency invest in disconnected, application-first pilots. Each pilot adds another point-to-point integration, making the eventual cost of connection higher than the innovation itself. The result is an ever-growing portfolio of "successful" pilots that never scale.

Ignoring operational reality: Administrative focus remains on documenting official system workflows while ignoring the informal workarounds (unregulated spreadsheets, manual data transfers) that staff actually use. AI implementations built on the official workflow fail in practice because they are trained on a fiction.

Undersized technical support: While 40% of IT leaders report staff are unequipped for AI adoption, leadership frequently underestimates the need for specialized roles (information architects, data governance leads) whose absence ensures that AI tools operate on unreliable data.

The Role of HHS

HHS should prioritize programs that address this fluency gap directly. Profession-specific AI competency frameworks would ensure that decision-makers understand not just the potential of AI, but the infrastructure, governance, and organizational change it requires. Programmatic design should reward organizations that demonstrate operational readiness—the ability to produce a complete audit trail and documented alignment between clinical and technical teams—before allocating federal capital to AI implementation.

Question #8: *Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care?*

Interoperability is the single highest-leverage intervention for accelerating clinical AI. The current landscape of fragmented “digital islands” forces every AI implementation to solve the same data reconciliation problem independently—consuming resources that should go toward innovation.

Widening Market Opportunities

When data flows through standardized interfaces, specialized AI tools can be deployed across different health systems without the prohibitive cost of custom integrations. This shifts the market from monolithic vendor platforms to a modular ecosystem where best-in-class tools compete on merit. Standardized APIs (including FHIR for clinical data exchange and context-aware standards like NGSI-LD for metadata-rich environments) lower the barrier to entry for smaller innovators, enabling startups to plug into enterprise health systems without rebuilding data pipelines from scratch.

Fueling Research

Interoperability transforms research from retrospective data reconstruction to real-time, high-fidelity analysis. Linking disparate signals (genomic markers, environmental exposures, symptoms) into unified data models enables rare disease diagnostics that are impossible in siloed databases. Standardized data collection (using CDISC standards for clinical trials, FHIR for clinical data) ensures real-world data is born ready for analysis. Federated learning models—where AI is trained across multiple institutions without moving raw patient data—depend entirely on interoperable infrastructure to function.

Critical Priorities

Category	Key Priority
Data Types	Multimodal integration: combining EHRs, claims, genomic data, and real-time biometric data from wearables into unified patient models.
Data Standards	FHIR for clinical exchange; context-rich standards (NGSI-LD, JSON-LD) for metadata-intensive environments where provenance and accuracy are first-class attributes.
Benchmarking	Agent-oriented benchmarks (e.g., MedAgentBench) that measure AI's ability to perform complex, multi-step clinical tasks—not just medical knowledge tests.
Data Quality	Data-quality-at-source protocols: ensuring every data point is findable, accessible, interoperable, and reusable (FAIR) at the moment of creation.

Question #10: *Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?*

HHS should prioritize research that bridges the gap between symbolic reasoning (ontologies, structured knowledge) and probabilistic modeling (large language models). Published findings consistently show that clinical AI adoption fails when it lacks the deterministic validation layer required to prevent hallucinations and performance degradation in high-stakes environments.

1. Ontological Harmonization

The primary bottleneck in clinical AI is not model capability but data fragmentation. Research into hybrid mapping systems—coupling classical lexical matchers with LLMs—has shown substantial improvements in aligning complex medical ontologies. HHS should fund research into scalable methods for resolving semantic inconsistencies across EHR systems, claims databases, and research repositories, ensuring that AI models reason from consistent definitions rather than conflicting data.

2. Agentic AI with Deterministic Grounding

Research should advance AI systems that autonomously plan and execute clinical tasks (agentic AI) while remaining anchored in verifiable data. The ARPA-H ADVOCATE program offers a promising model for "supervisory agents" that ensure patient-facing AI remains safe and aligned with clinical guidelines. Additional priorities include transforming clinical practice guidelines into machine-readable constraints and developing uncertainty quantification methods for real-time physiological data.

3. Reducing the Data Preparation Burden

Research consistently finds that the majority of data scientist and researcher time is consumed by data preparation rather than analysis.[1] HHS should fund research into "data-quality-at-source" protocols that ensure clinical data is standardized, labeled, and ready for AI consumption at the moment of creation—eliminating the retrospective data reconstruction that currently drains research productivity. Equally important is research into preventing "model collapse," where AI models degrade when trained on recursively generated synthetic data rather than high-fidelity, curated records.

Published Evidence on Adopted AI Tools (Questions 10a–10b)

The published literature on clinical AI economics has matured significantly. A systematic review of economic evaluations found that 98% of studies concluded AI-based strategies were cost-effective or cost-saving,[10] with projections suggesting annual net savings of $200–$360 billion in U.S. healthcare (a 5–10% reduction in total spending).[11] However, fewer than half of studies explicitly report their costing methods, and only 28% include implementation costs—indicating that the full financial burden of AI adoption is systematically underestimated in the literature.

The literature also reveals that trust in medical AI flows hierarchically: from digital literacy to the AI system, and only then to the physician and institution. This reinforces the importance of the physician's intermediary role in building institutional trust—and the danger of deploying AI in ways that bypass clinical judgment.

Closing

We appreciate HHS's commitment to gathering concrete, experience-based input on clinical AI adoption. The consistent theme across our responses is that the barriers to AI in healthcare are primarily infrastructural and organizational—not algorithmic. Investing in data infrastructure, operational readiness, and executive digital fluency will yield greater returns than investing in AI models alone. We welcome the opportunity to engage further with HHS on these priorities.

Respectfully submitted,

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References

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[10] PMC systematic review (PMC12795340): 98% of economic evaluations found AI cost-effective or cost-saving.

[11] Medrxiv preprint 2025. Projected annual net savings of $200B–$360B (5–10% reduction in total U.S. healthcare spending).